

MCMINNVILLE, OR, USA

WMO: 726881

Lat: 45.195N

Lon: 123.134W

Elev: 159

StdP: 14.61

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 94273

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 25.8	(c) 28.2	(d) 12.8	(e) 10.6	(f) 30.1	(g) 19.4	(h) 14.7	(i) 34.7	(j) 26.9	(k) 49.7	(l) 24.4	(m) 48.9	(n) 3.9	(o) 350	(p) 0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 28.8	(c) 92.8	(d) 67.0	(e) 88.9	(f) 66.1	(g) 84.4	(h) 64.7	(i) 68.9	(j) 87.6	(k) 67.3	(l) 85.7	(m) 65.7	(n) 82.7	(o) 7.9	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	84.0	73.8	60.7	79.6	72.5	58.9	74.8	70.9	33.0	87.2	31.6	85.7	30.4	82.8	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	17.4	15.3	DB	20.0	100.2	5.2	3.3	16.3	102.6	13.3	104.5	10.4	106.4	6.6	108.9
			WB	18.8	73.6	5.0	3.3	15.2	75.9	12.2	77.9	9.4	79.8	5.7	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	41.5	43.5	46.7	50.2	56.4	61.5	67.9	68.2	63.0	53.7	45.9	40.8
	DBStd	11.09	5.90	5.13	4.79	5.38	5.92	5.74	5.01	4.97	5.73	5.40	5.92	6.32
	HDD50	1116	267	188	127	60	10	0	0	0	24	149	291	
	HDD65	4580	728	602	569	445	275	137	23	23	105	351	573	749
	CDD50	2332	4	5	23	66	208	344	556	563	390	140	26	7
	CDD65	322	0	0	0	1	8	32	114	121	44	2	0	0
	CDH74	5052	0	0	0	37	215	580	1738	1683	762	37	0	0
	CDH80	2071	0	0	0	5	57	225	761	735	282	6	0	0
(14) Wind	WSAvg	5.7	5.9	6.4	6.4	6.0	5.4	5.6	5.7	5.3	5.0	5.1	5.8	6.0
(15) Precipitation	PrecAvg	42.8	6.9	5.0	4.6	2.8	2.0	1.0	0.4	0.6	1.5	3.1	6.3	7.6
	PrecMax	77.4	13.7	13.2	8.4	7.0	4.3	2.7	2.2	4.6	7.6	7.2	12.7	20.0
	PrecMin	24.3	0.4	0.4	0.8	0.3	0.0	0.0	0.0	0.0	0.0	0.2	0.6	0.2
	PrecStd	9.9	3.6	3.1	1.9	1.6	1.1	0.7	0.5	0.9	1.5	1.8	3.2	4.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	60.8	69.5	79.2	86.6	93.5	98.8	97.4	93.0	79.1	64.1	61.6
		MCWB	54.2	50.5	56.2	61.8	65.2	67.6	69.5	67.7	66.1	58.9	58.7	55.8
	2%	DB	54.5	56.8	63.4	72.1	80.4	86.3	92.6	92.5	87.7	72.4	60.2	55.6
		MCWB	51.0	49.4	53.5	57.6	62.7	65.4	67.2	66.9	64.1	57.6	55.6	52.5
	5%	DB	52.2	54.1	59.1	65.7	74.5	80.8	88.2	88.3	82.4	67.8	57.1	52.2
		MCWB	49.3	48.8	51.2	54.3	60.2	63.3	66.6	66.1	62.8	56.2	52.9	49.5
	10%	DB	50.0	51.9	55.5	61.2	69.7	75.1	83.7	83.5	78.0	63.6	54.8	49.4
		MCWB	47.3	47.4	49.8	52.2	58.2	60.9	65.1	64.9	61.3	55.7	51.3	46.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	54.0	57.9	63.4	67.6	69.9	72.3	70.6	69.0	63.9	60.3	56.3
		MCDB	58.0	56.9	65.7	77.6	83.1	89.0	93.5	90.8	86.1	72.6	62.9	60.5
	2%	WB	52.0	51.6	54.8	58.8	64.1	66.4	69.0	68.4	66.1	60.5	56.8	53.1
		MCDB	53.6	54.8	60.5	69.8	77.0	83.7	87.8	87.6	82.3	67.2	59.6	55.4
	5%	WB	49.8	49.7	52.7	55.4	61.3	64.0	67.1	66.8	64.2	58.5	53.9	49.9
		MCDB	51.8	52.7	57.8	63.7	72.5	78.8	86.2	85.1	78.5	64.4	56.1	52.0
	10%	WB	47.6	47.7	50.6	53.0	58.8	61.8	65.3	65.2	62.3	56.8	52.0	46.8
		MCDB	49.4	51.1	55.2	59.5	68.6	74.1	82.7	82.1	74.8	62.2	54.4	48.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	13.6	16.3	19.2	21.5	23.3	28.8	28.8	26.5	18.9	13.0	10.2
		MCDBR	12.6	17.3	23.2	29.6	32.1	33.8	37.0	37.2	35.7	27.8	15.1	11.8
	5% WB	MCWBR	9.5	11.7	14.7	17.2	17.5	15.8	15.1	15.0	15.9	15.3	10.4	9.2
		MCWBR	11.3	13.8	20.1	25.7	28.7	30.2	34.2	33.1	31.1	20.0	11.5	11.1
(40) Clear-Sky Solar Irradiance	taub		0.322	0.319	0.331	0.359	0.370	0.362	0.352	0.355	0.345	0.334	0.327	0.328
		taud	2.518	2.535	2.504	2.392	2.378	2.411	2.460	2.467	2.488	2.527	2.510	2.463
	Ebn at Noon		252	277	287	286	285	287	288	284	279	266	247	234
		Edh at Noon	21	25	29	36	37	36	34	33	30	25	21	21
(44) All-Sky Solar Radiation	RadAvg		384	646	942	1367	1740	1929	2202	1881	1407	819	443	312
	RadStd		61	89	122	124	170	177	147	122	121	100	60	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+47	

Nomenclature: See separate page